

Gentle by knowing: how much must a care robot know about a person to lift their arm safely?

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3 **ABSTRACT**

4 Robots moving a person's limb are protected mainly by contact-force limits. We asked how much
5 a robot must know about a person to lift their forearm without pressing a joint into the end of its
6 range, and whether force sensing can replace that knowledge. In a musculoskeletal simulation of
7 a robot supporting the forearm from below, 40% of simulated persons had a restricted shoulder,
8 30% a restricted elbow. Controllers used no knowledge (a fixed lift), a categorical limitation flag,
9 recorded joint ranges with realistic error (feeding a population body model or a policy learned
10 offline in simulation), or perfect knowledge. Hypotheses were specified in a dated protocol before
11 each round and tested on new simulated persons; post hoc analyses are labelled. A force-
12 threshold controller never reduced harmful outcomes by significantly more than 5 percentage
13 points; its best setting cut harm by 4–7 points at the cost of 40–51% of safe completions, because
14 95–100% of harmful outcomes began during the first attempt. Knowledge helped mainly by telling
15 the robot whom not to lift. The learned policy reduced harmful outcomes from 110 to 25 of 300
16 persons at similar safe completions (174 vs. 177), using a care record with 20° error. It replicated
17 and held under several degradations, but shrank to about two fifths when half of the restrictions
18 were unrecorded. Post hoc, knowing which joint was restricted, plus the settled posture, sufficed.
19 Gentleness came from knowing the person before contact and being willing to decline, not from
20 sensing force.

21 **Keywords:** physical human–robot interaction, caregiving robots, gentle handling, safety, musculoskeletal simulation, joint contracture,
22 range of motion, abstention

1 INTRODUCTION

23 Moving a person's limb is among the most frequent actions in care: an arm is lifted to wash the axilla,
24 to pass a sleeve, or to reposition the person in bed, and it is a natural target for assistive robots (Chow
25 and Kemp, 2016; Koh and Qureshi, 2025). Many care recipients cannot move freely. About half of
26 adults admitted with stroke develop at least one contracture within six months (Kwah et al., 2012), and
27 contractures are common among residents of long-term care (Wagner et al., 2008). In long-term care,
28 contractures independently predict minimal-trauma fractures (Lam et al., 2016), some of which occur

29 during diaper changes, repositioning or transfers (Hagino et al., 2022); in hemiplegia, the way the arm is
30 moved is associated with shoulder pain (Kumar et al., 1990; Wanklyn et al., 1996).

31 Collaborative-robot safety rests largely on limits on contact force and pressure (International Organization
32 for Standardization, 2016), derived from pain-onset thresholds. Adjacent standards ask more: wearable
33 assistive robots may apply assist force only within the user's passive range of motion (Japanese Industrial
34 Standards Committee, 2016), rehabilitation robots must limit movement to pre-set, patient-specific limits
35 (International Electrotechnical Commission, 2019; Massardi et al., 2023), and personal-care robots are
36 expected to seek external assistance or stop when decision uncertainty remains unacceptable (International
37 Organization for Standardization, 2014). For end-effector limb handling, it is not known which of these
38 measures matter, nor how much the robot must know about the person.

39 One intuitive answer is that a sensitive force limit makes knowledge unnecessary; another, that the robot
40 needs a detailed personal model of the body. Human carers suggest a third: they read in the chart that “the
41 left shoulder is stiff” and handle the arm differently or ask for help. We make the robot's knowledge of the
42 person the independent variable and measure gentleness as the absence of load pressed into the end of a
43 joint's range.

44 Related systems reposition passive limbs within age-group joint limits and pause when no valid pose
45 exists (Koh and Qureshi, 2025), plan dressing within each participant's measured range (Kapusta et al.,
46 2019), or predict functional range from clinical scores (Liu et al., 2025); their outcomes are task success,
47 realism, tracking or strain, not harm across levels of information about the person. Peiros et al. (2023)
48 argued that the same external force can correspond to very different joint loads, and Bessler et al. (2021)
49 that industrial force limits do not transfer directly to patients; yet in joint-aligned single-degree-of-freedom
50 devices, torque thresholds are used to find the end of range without reported safety problems (Zhang et al.,
51 2002; Lee et al., 2025). Our claim is therefore scoped to end-effector support of a multi-joint limb, where
52 the loaded joint is not on the sensor's axis.

53 Comparing a baseline, an inference model and an omniscient planner has precedent in autonomous
54 driving (Sunberg et al., 2017), and assistive robotics compares generic and personalised human models
55 (Qin et al., 2025; Christou et al., 2025; Yang et al., 2026). To our knowledge, no prior work varies the
56 amount of person-specific body information available to a physically assistive robot—from none, through a
57 categorical record flag and documented joint ranges of varying accuracy, to a body model, a learned policy
58 and an oracle—and reports harm and safe task completion at each level with declining as an explicit option,
59 or tests whether contact-force sensing can substitute for that information. We contribute:

- 60 1. Evidence that force-threshold control does not substitute for this knowledge in end-effector support of
61 a multi-joint limb, with the mechanism: harm begins in the first attempt, before a guard can switch
62 strategy (Figure 2).
- 63 2. An *information ladder* of harm and safe completions at seven levels of knowledge (Figure 3), in
64 which a policy learned offline in simulation cuts harm to about a quarter of a fixed lift's at similar safe
65 completions; it survives several degradations but loses much of its benefit when the record misses
66 restrictions (Figure 4).
- 67 3. A post hoc analysis of what the robot needs to know—which joint is restricted, not by how many
68 degrees—and an account of what did not help, with every hypothesis reported (Supplementary
69 Table S1).

2 MATERIALS AND METHODS

2.1 Simulated body and population

We used the MyoSuite myoArm model (Caggiano et al., 2022) (MyoSuite 2.12.2) in MuJoCo 3.6 (Todorov et al., 2012). The humerus was not welded to the thorax, so the arm hangs from the shoulder complex. Muscles were driven by a stretch reflex with muscle velocity and length feedback, following the structure of Yu et al. (2025), with a 25-ms afferent delay (our choice) and a tonic baseline activation of 0.01; there is no voluntary activity, so the simulated person neither assists nor resists. Muscle strength, passive muscle force and joint damping varied log-normally across persons ($\sigma = 0.25$; segment masses and inertias $\sigma = 0.175$; Supplementary Table S3).

Each person independently had a restricted shoulder elevation with probability 0.4 and a restricted elbow flexion with probability 0.3, in line with the contracture rates above. A restricted joint could move a free range drawn from $U(4^\circ, 16^\circ)$ (shoulder) or $U(0^\circ, 10^\circ)$ (elbow) above the person's own settled posture before its joint limit engaged. The free ranges were chosen, as stated in the protocol, to overlap the joint excursions produced by the task (about 7° – 14° at the shoulder and up to 9° at the elbow), which makes restrictions consequential; absolute harm rates therefore reflect the population design, and we interpret only paired differences between controllers. A *shifted* population had restriction probabilities 0.6 (shoulder, $U(2^\circ, 12^\circ)$) and 0.5 (elbow, $U(0^\circ, 8^\circ)$). By default the joint limit is hard; in *graded end-feel* conditions, resistance ramps up over a window of width $w \in \{6^\circ, 12^\circ, 20^\circ, 25^\circ\}$ centred on the limit (our derivation from passive-resistance measurements, Vetter et al., 2023; Novotny et al., 2000; Rao et al., 2021).

2.2 Robot and task

The end effector is a position-servoed capsule with three translational and three rotational joints (effectively five degrees of freedom, since the capsule is axisymmetric; 6000 N/m, 20 N·m/rad, 50 Hz) that approaches the forearm from below. It raises the forearm centre by 5 cm, closing the loop on the observed forearm height, and holds for 0.5 s. A *strategy* is a support site along the forearm (8 sites, 3 cm apart), an approach angle ($-40^\circ, -20^\circ, 0^\circ$) and a lifting speed (0.10 or 0.25 m/s), giving 48 strategies. Proximal support elevates mainly the shoulder; distal support also flexes the elbow (Figure 1). No single strategy is safe for everyone: proximal support overloads a restricted shoulder, distal support a restricted elbow.

2.3 Gentleness: harm and outcomes

Harm is the load the robot presses into the shoulder-elevation and elbow-flexion limits: the peak of the 50-ms moving average of the joint-limit constraint torque (excluding the first 100 ms), summed over the two joints. From round 23 onward, passive limit torque present before contact was subtracted; earlier rounds used the gross torque. An outcome is *harmful* if $\text{harm} \geq 0.5 \text{ N}\cdot\text{m}$ (primary threshold). We also report $2.5 \text{ N}\cdot\text{m}$, close to the mean torque ($2.7 \pm 1.1 \text{ N}\cdot\text{m}$) that therapists applied to reach passive elbow end range in children after fracture (Eastman et al., 2022), a threshold sweep, and a threshold-free mean load. A *safe completion* is a completed lift with harm below the threshold; a declined lift counts as neither. Harm is never shown to any controller except the oracle. We regard it as a mechanical exposure proxy, not as pain or injury.

2.4 What the robot may know

The *care record* gives each joint's free range with Gaussian error of standard deviation $\varepsilon \in \{4^\circ, 10^\circ, 20^\circ\}$; reported standard errors of measurement of 14 – 25° between raters for symptomatic shoulders (Hayes et al.,

2001) and 6–10° within a session for the elbow after stroke (Lin et al., 2020) make 20° our realistic default. Unrestricted joints are recorded as unrestricted, so the record also tells which joints are restricted. Real records are often categorical, such as the MDS 3.0 upper-extremity item, which codes shoulder, elbow, wrist and hand together as 0/1/2 (Centers for Medicare & Medicaid Services, 2025), and often incomplete: in one nursing-home study, reduced range of motion was found in 69% of residents but documented for 4% (Austen et al., 2026). We therefore also consider a categorical limitation flag (shoulder and elbow merged) and a *half-documented* record in which each restricted person is unrecorded with probability 0.5. The robot observes the person's settled shoulder and elbow angles before contact and its own wrist force during the lift.

2.5 Controllers

Controllers differ only in what they know and whether they may decline.

F (fixed rule). Among strategies completing $\geq 90\%$ of 60 training persons, the one with the lowest median harm.

R (reactive force guard). Runs F; if the wrist force exceeds m times the 95th percentile of the force in harmless training lifts ($m \in \{0.6, 0.8, 1.0, 1.2\}$), it stops, retracts for 0.5 s and tries the best strategy supporting at least 9 cm away (which loads the other joint), up to three attempts.

CAT, CAT_{half}. Decline if the (complete or half-documented) categorical flag marks a restriction; otherwise F.

S (record threshold). Decline if the recorded free range of either joint is below 3.5° (locked on 280 earlier persons for 4° records); otherwise F.

M (population body model). A library of 199 simulated unrestricted bodies, each run with all 48 strategies, predicts each joint's excursion and each strategy's completion rate. M keeps F if F is predicted to stay at least 3° inside both recorded ranges; otherwise it uses the margin-keeping strategy with the highest predicted completion (at least 0.8); if none keeps the margin, it executes the least-overshooting strategy if its overshoot is at most 3°, and declines otherwise.

L (policy learned offline in simulation). Training data are 400 simulated persons from earlier rounds, each with outcomes for all 48 strategies. From six features—the two recorded free ranges (clipped at 30°), two unrestricted flags, and the two settled joint angles—a k -nearest-neighbour model ($K = 25$, standardised features) estimates each strategy's probability of a safe completion among the 25 most similar training persons. L picks the best strategy and declines if its estimate is below $\tau = 0.65$, the training-set value with the fewest harmful outcomes among those keeping safe completions within 2.5 percentage points of F. Persons recorded as unrestricted receive F. A variant without posture input (four features, $\tau = 0.60$) was specified in advance for round 25.

O (oracle). Knows the outcome of every strategy; picks a safe completion if one exists, otherwise declines (a ceiling, not an achievable controller).

2.6 Protocol and statistics

The study proceeded in confirmation rounds 17–26 (Supplementary Table S6). Before each round's data were generated, we wrote the hypotheses, controllers, thresholds, sample size, margins and analysis into a dated protocol document; the document was kept locally and not externally registered. Every confirmation set consists of simulated persons (random seeds) never used before.

All comparisons are paired (same simulated persons). We report paired differences in harmful-outcome and safe-completion rates with 95% confidence intervals (paired t); the preregistered analysis code used

151 $t = 2.1$ for all $n > 20$, which makes the intervals about 6% wider than exact (conservative). A harm
152 hypothesis was supported if the upper confidence bound of the difference was below zero; for force guards,
153 the criterion was an upper bound below -5 percentage points. Non-inferiority on safe completions used
154 margins fixed in advance (2–2.5 percentage points, or an absolute count stated per round); equivalence
155 used ± 0.05 . We did not correct for multiplicity. Analyses marked *post hoc* were not preregistered; their
156 baselines were tuned on the 400 training persons only, with the criterion used for τ , and evaluated on the
157 600 confirmation persons of rounds 24 and 25 that have complete outcome grids.

158 *Deviations* (all listed in Supplementary Table S2). An implementation bug was found after round-17
159 confirmation data had been generated; the body-model arm was re-run on the same 40 persons after the
160 other arms had been seen. Round 22 counted resting passive tension under wide end-feel as harm; a
161 round-23 re-run of 12° with the net measure reached the same conclusions. After round 18R, the primary
162 outcome changed from paired mean harm to the harmful-outcome rate with a non-inferiority condition
163 on safe completions, because declining trivially minimises mean harm. Persons from earlier confirmation
164 rounds were reused as development or training data, never as confirmation data.

3 RESULTS

165 3.1 Force sensing does not substitute for knowledge

166 The fixed rule F harmed 34–40% of persons in every hard-limit confirmation set (e.g. 229 of 600 in round
167 22). Across four guard thresholds, under a hard limit and graded end-feel of 6, 12, 20 and 25° , no force
168 guard reduced the harmful-outcome rate by significantly more than the preregistered 5 percentage points
169 (Figure 2). This is a failure to show a large reduction, not proof of none: the tightest guard produced small
170 significant reductions at 6° (-4 points), 20° (-4) and 25° (-7 [-11 , -2]), but not with a hard limit or
171 at 12° , and each of these cost 40–51% of the safe completions. With a hard limit, tightening the guard
172 reduced safe completions from 69 to 40 of 120 persons at an unchanged harm rate. The mechanism was
173 timing: at the primary threshold, 95–100% of harmful outcomes occurred during the first attempt, before
174 the controller could switch strategy. It was not a lack of observability. With 12° end-feel, the wrist force
175 tracked joint load within a person (median Spearman correlation 0.75; 0.10–0.35 with a hard limit or 6°),
176 yet the guard still acted after the joint had been loaded.

177 3.2 The information ladder

178 Knowledge helped mainly by telling the robot whom not to lift. The body-model rule M, which declines
179 when it predicts no safe lift, reduced the harmful-outcome rate by 14–26 percentage points in every
180 confirmation condition in which it was tested (rounds 21–26; e.g. -0.19 [-0.23 , -0.16], 113 vs. 229 of
181 600) with safe completions comparable to F (351 vs. 344); without its decline option it was no better than
182 F (Section 3.5).

183 Figure 3 gives harmful outcomes and safe completions for 300 new persons at each level of knowledge
184 (counts in Supplementary Table S4). A complete categorical flag used to decline removed all harm but
185 also 37% of safe completions; with half the flags missing it left 60 harmful outcomes. Joint ranges with
186 20° error combined with the record threshold S or the body model M reduced harm by 39–54%. The
187 offline-learned policy L reduced harmful outcomes to 25, fewer than M (-0.14 [-0.19 , -0.09]), with
188 more safe completions than M ($+0.04$ [0.00 , $+0.08$]) and similar safe completions to F (174 vs. 177;
189 preregistered non-inferiority not established, difference -0.01 [-0.05 , $+0.03$]). Record accuracy mattered
190 little once L was used (4° : 184 safe, 20 harmful; 20° : 174, 25).

The remaining distance to perfect knowledge (237 safe, 0 harmful) had two parts. Of the 63 safe completions L missed, 47 (75%) were unnecessary refusals of persons who had a safe strategy. Of L's 25 harmful outcomes, 18 were persons for whom no safe strategy existed, whom L should have declined. In threshold-free terms, the mean robot-caused end-range load per person was 6.73 N·m under F, 3.19 under M and 0.79 under L ($L - F: -5.94 [-7.78, -4.10]$). The advantage is not only a matter of lifting fewer people: post hoc, on the 423 of 600 persons L lifted (rounds 24–25), L harmed 53 and F would have harmed 95 ($-0.10 [-0.13, -0.07]$).

3.3 Robustness of the learned policy

On further new persons (Figure 4; counts in Supplementary Table S5), L replicated its reduction relative to F ($-0.27 [-0.32, -0.21]$) and to M ($-0.08 [-0.12, -0.03]$), but its preregistered safe-completion non-inferiority to M did not replicate ($-0.01 [-0.05, +0.03]$). The reduction relative to F and M held with Gaussian noise (SD 5°) on the observed posture or without posture input, in the shifted population (170 vs. 38 harmful of 250, without retraining), and under 20° graded end-feel, with safe completions never more than 1.3 percentage points below F's.

Record completeness mattered more. With half of the restrictions missing from the record, L still harmed fewer persons than F (64 vs. 108) and M (81), but its harm was 59% of F's rather than about a quarter, and a categorical flag used to decline (CAT_{half}) harmed fewer persons than L (50 vs. 64; $+0.05 [+0.02, +0.07]$) at 29 fewer safe completions. A record-based controller treats a person whose restriction is not recorded like the fixed rule: the 85 such persons were harmed at exactly the fixed-rule rate (50 vs. 50).

With test physics differing from training—three-fold stretch-reflex gains or wider individual variation ($\sigma = 0.40$)—L's reduction relative to F and M held, with safe completions 3.3 and 0.3 percentage points below F's. These perturbations kept the joint-limit geometry, and F's harm rate barely moved (36% and 40%, vs. 36–37% in the standard world), so they test muscle dynamics and body variation, not a different body structure.

Three checks specified in advance (rounds 24–25, 600 persons) found no hidden artefact. In hindsight, the best of all 48 fixed strategies harmed 203 persons, but with only 135 safe completions; the preregistered F harmed 218 (rank 9 of 48); L harmed 53. On the persons L lifted, glenohumeral shear did not increase (paired mean difference $L - F -3.6$ N, median 0). When records over-estimated the free range of restricted joints by 5° or 10° , as with worsening contractures, L's harmful outcomes rose from 53 to 57 and 68, whereas M's rose from 118 to 160 and 210. In a post hoc sweep of the harm threshold from 0.25 to 5 N·m (Supplementary Figure S1), L had the fewest harmful outcomes among F, S, M and L at every threshold, but F completed more lifts safely at almost every threshold. In the preregistered secondary analysis at 2.5 N·m, L reduced harm in every confirmation set but completed 4–8 percentage points fewer lifts than F.

3.4 What the robot needs to know (post hoc)

With 20° error, recorded degrees say little about a free range of a few degrees. On the 600 confirmation persons with complete grids, we compared L with baselines tuned on the training persons only (Supplementary Figure S2). A nearest-neighbour policy given only the two restriction flags and the settled posture (no recorded degrees; $\tau = 0.65$ by the same criterion) performed like L: 353 safe and 46 harmful vs. 341 and 53 ($L - \text{flags: harmful } +0.01 [-0.01, +0.03]$, safe $-0.02 [-0.04, 0.00]$). A lookup table choosing the best training strategy per restriction type had only four operating points; at the tuned setting it completed more lifts (396) at a higher harm (94). Re-tuned for 20° records, M harmed 113 persons at 335 safe completions

($L - M$: $-0.10 [-0.13, -0.07]$), and S could not meet the completion criterion at any setting. Across the whole trade-off, the learned policies lay below the record-threshold and body-model curves.

3.5 What did not help

Choosing the support site to route motion to a joint with spare range, without declining, did not reduce harm in round 21 ($-0.03 [-0.09, +0.02]$; preregistered null). Identifying the individual body by two gentle probe lifts was supported in round 18 ($n = 40$) but did not replicate with 120 persons; in round 19 the probes loaded a restricted joint beyond $0.5 \text{ N}\cdot\text{m}$ in 18 of 71 probed persons, and the probing controller harmed more persons than the same rule without probing (29 vs. 21 of 120). In round 19, the body model with declining harmed fewer persons than the reactive guard (21 vs. 42 of 120) but completed fewer lifts safely (66 vs. 76); the learned policy was not compared with the guard directly.

4 DISCUSSION

Why knowledge, and not force.

In end-effector support of a multi-joint limb, the loaded joint is not the one the force sensor sees, and the load builds within the first attempt. A guard can only react after the joint has been loaded; tightening it trades completed tasks for a small reduction in harm. Knowledge acts before contact, and the decision that mattered most was whether to lift at all. Force limits remain necessary for contact safety; our results argue that they are not sufficient for joints.

What kind of knowledge.

The robot did not need a detailed personal model: probing added harm, and record accuracy, posture input and even the recorded degrees were dispensable. What mattered was knowing *which* joint is restricted, and having learned, from the simulated outcomes of many persons like this one, which lifts are safe and when to decline. M also relies on a precomputed library, but one of unrestricted bodies only, from which it must infer harm through predicted excursions and record margins; L learns harm labels directly from simulated persons of the restricted population. The shifted-population test suggests that this is not mere distribution matching. In this sense, the value of a simulator as a world model lay in generating outcome labels from which a decision policy can be learned offline.

The cost of declining.

Declining is the dominant mechanism, and it is not free: it hands the task to a human carer, whose handling carries its own risk (all harm counts here are robot-caused), or delays it. L declined 30% of persons, the oracle 21%. Because τ was chosen to keep safe completions close to the fixed rule's, similar completion counts are partly a design choice (Supplementary Figures S1 and S2 show the trade-off); a deployed system would choose τ with the care team, according to the availability of a human alternative. For human carers, the assessment components of patient-handling programmes have not been evaluated separately (Teeple et al., 2017), and for falls, clinical judgement was not inferior to a scored risk tool (Meyer et al., 2009); our ladder describes robots, not carers. Our simulated persons do not wince or speak; our results apply most directly to care recipients who cannot give such cues reliably, such as people with advanced dementia or aphasia.

268 **Implications for design.**

269 Robots that move limbs by end-effector contact should (i) consult joint-level information about the person
270 before contact—at least which joints are restricted, which is coarser than a goniometric range but finer
271 than merged-limb flags such as the MDS upper-extremity item; (ii) include an explicit option to decline
272 and hand over the task; and (iii) not rely on force thresholds to protect joints. The first two echo adjacent
273 standards (Japanese Industrial Standards Committee, 2016; International Electrotechnical Commission,
274 2019; International Organization for Standardization, 2014); we add quantitative evidence of their value.
275 Because unrecorded restrictions cannot be protected, the completeness of the record matters more than its
276 precision.

277 **Limitations.**

278 All results are simulated; no robot or human data were collected. Training and test simulators share the
279 same musculoskeletal structure and joint-limit geometry; a real person differs in more ways, including
280 voluntary resistance, pain responses, soft-tissue contact, skin friction and body geometry. The harm
281 measure is a mechanical exposure proxy without a dose–response, and the peak constraint torque depends
282 on the simulator’s constraint model, which we did not vary. Independent shoulder and elbow restriction
283 and uniform free ranges, chosen to interact with the task, are assumptions, so absolute harm rates are not
284 predictions for real populations. We studied one task, one end effector and a passive recipient, and only a
285 force-based reactive controller (not, e.g., a kinematic guard). The protocol was dated and kept locally rather
286 than registered externally, earlier confirmation persons were reused for development, and the analyses of
287 what the robot needs to know are post hoc. Next steps are an arm phantom with adjustable joint limits and
288 changed body geometry between training and test.

289 **Conclusion.**

290 For a robot lifting a person’s arm, gentleness depended less on sensing the contact than on knowing,
291 before contact, which joint is restricted and when not to lift. Force thresholds could not substitute for that
292 knowledge, and unrecorded restrictions remained unprotected.

CONFLICT OF INTEREST STATEMENT

293 The author(s) declare that the research was conducted in the absence of any commercial or financial
294 relationships that could be construed as a potential conflict of interest.

GENERATIVE AI STATEMENT

295 [The author(s) declare that generative AI (Claude, Anthropic) was used in the creation of this manuscript:
296 to write and run simulation and analysis code, to search and summarise literature, and to draft and edit the
297 text, under the direction of the author(s). The author(s) reviewed and verified all code, results, references
298 and text and take full responsibility for the content.]

AUTHOR CONTRIBUTIONS

299 MW: conceptualization, methodology, software, formal analysis, investigation, writing – original draft,
300 writing – review and editing.

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SUPPLEMENTARY MATERIAL

Supplementary Table S1 lists every preregistered hypothesis with its outcome; Table S2 lists deviations and reuse of persons; Table S3 lists all simulation parameters with their source files; Tables S4 and S5 give the counts behind Figures 3 and 4; Table S6 lists the confirmation rounds. Supplementary Figure S1 shows the post hoc harm-threshold sweep and Figure S2 the post hoc trade-off between safe completions and harmful outcomes.

DATA AVAILABILITY STATEMENT

The simulation code, the dated protocol with all amendments, per-episode outcome data for every confirmation set, and scripts that regenerate every number, table and figure in this article are available at <https://github.com/maurice1128/gentle-by-knowing>.

ETHICS STATEMENT

This study used only computer simulation. No human participants, human data or animals were involved, and ethical approval was not required.

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FIGURE CAPTIONS

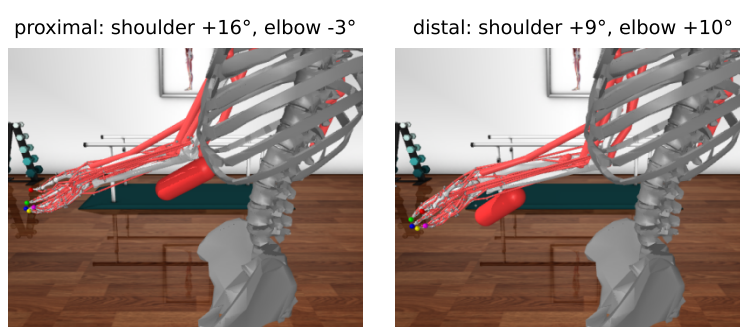


Figure 1. The task. A capsule supports the forearm from below and raises it 5 cm. The support site decides which joint moves (joint excursions shown for one unrestricted simulated person, seed 104): proximal support elevates mainly the shoulder, distal support also flexes the elbow. For a person with a restricted shoulder or elbow, the wrong site presses that joint into the end of its range.

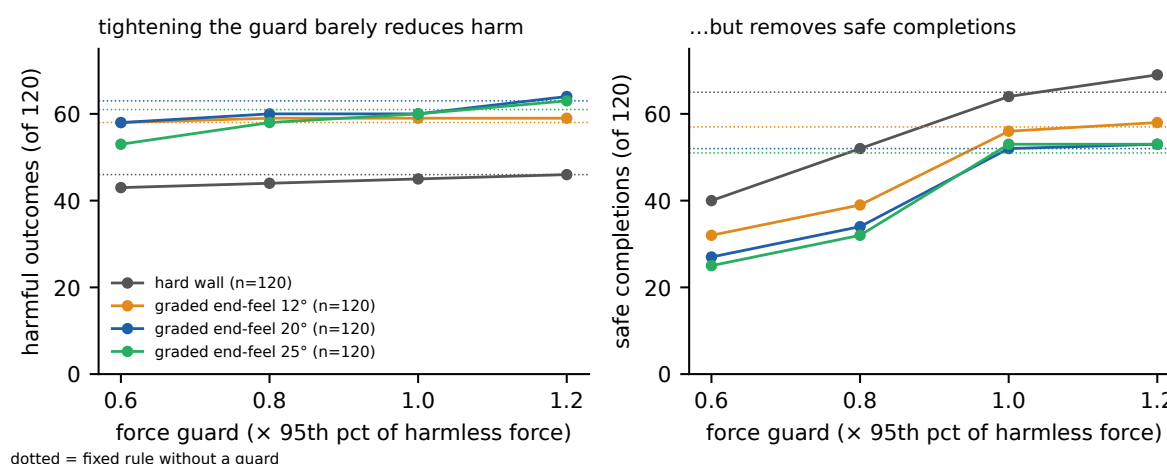


Figure 2. Force-threshold control at four thresholds: hard limit (round 21, 120 new persons) and graded end-feel of 12, 20 and 25° (round 23, the same 120 new persons under each width). Tightening the guard barely changes harmful outcomes (left) but removes safe completions (right). Dotted: fixed rule without a guard.

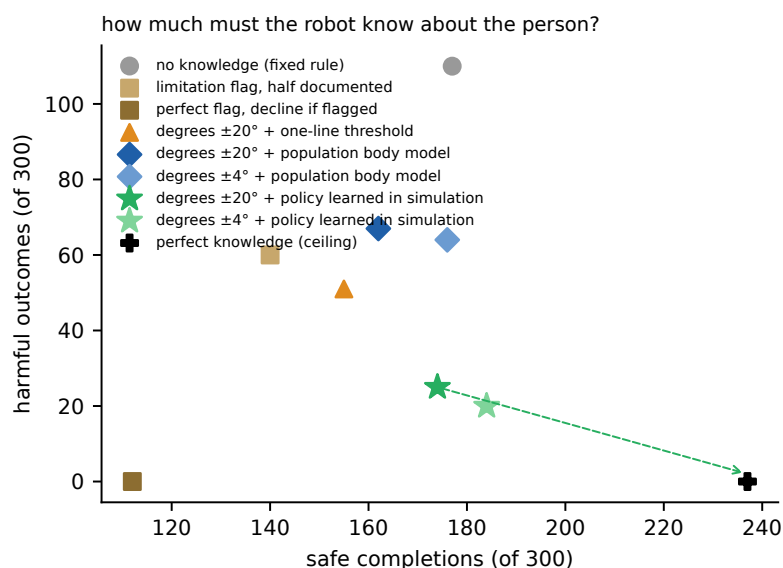


Figure 3. Information ladder on 300 new persons (round 24, harmful ≥ 0.5 N·m; lower right is better). Each marker is one level of knowledge about the person, from none (fixed rule F) to perfect knowledge (oracle O). The dashed arrow is the remaining gap from the learned policy to perfect knowledge. Counts, including declined persons, are in Supplementary Table S4.

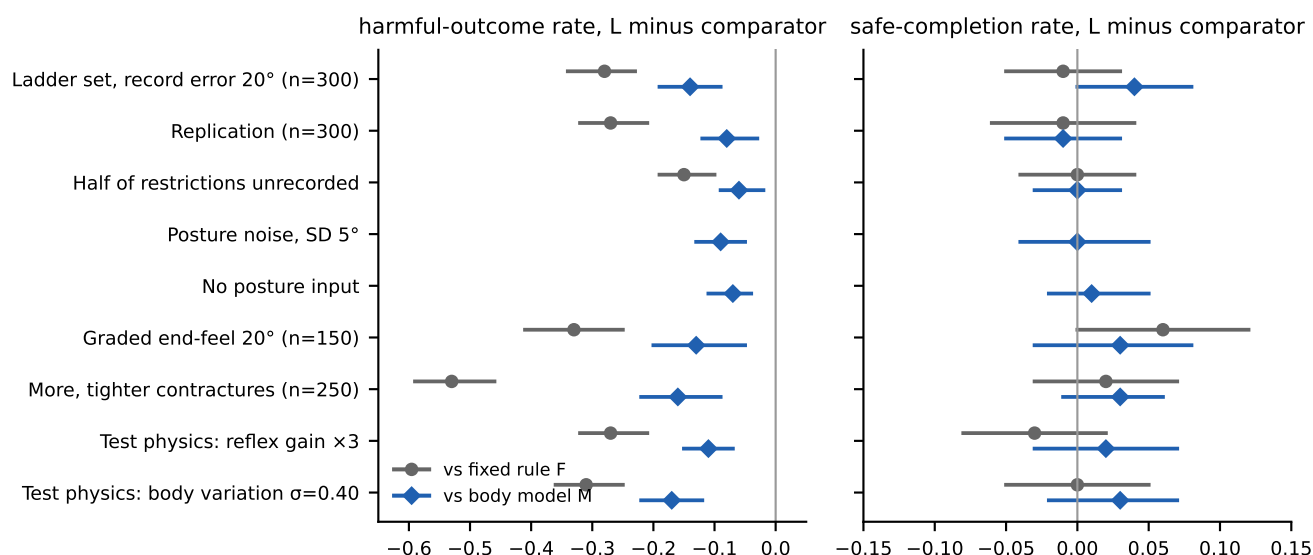


Figure 4. Robustness of the learned policy L. Paired differences (95% confidence intervals) in harmful-outcome rate (left) and safe-completion rate (right) between L and the fixed rule F (grey) or the body-model rule M (blue), in every preregistered confirmation condition (harmful ≥ 0.5 N·m). L, M and F were not retrained for any condition; the no-posture variant is the separately specified four-feature policy. The posture-estimate condition is Gaussian noise with SD 5° ; posture conditions were compared with M only, as preregistered. Counts are in Supplementary Table S5.